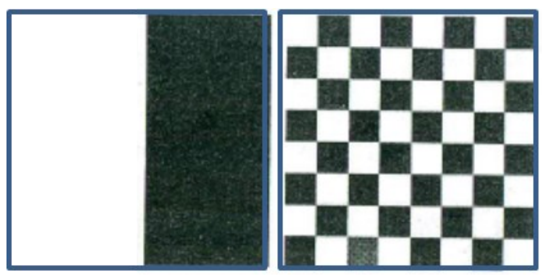
Homework #3

Due September 20th, Tuesday, before class starts

1. Two $N \times N$ images shown below are quite different, but their histograms are the same. Suppose that each image is blurred with a 3x3 averaging mask. Please ignore the low quality of the images. You can assume all the black regions have a uniform intensity of 0, and the white regions have a uniform intensity of 255. (20 pts)
   (a) Would the histogram of the blurred images still be equal? Explain.
   (b) If your answer is no, sketch the two histograms.
   (c) **Bonus question**: Discuss how the histogram of the blurred Figure1(b) is affected by different image size *N.* (5 pts)



(a) (b)

Figure1. (a) An image with white intensities on left and black intensities on the right. (b) A chessboard image.

Solution:

(a) The number of boundary points between the black and white regions is much larger in the image on the right. When the images are blurred, the boundary points will give rise to a larger number of different values for the image on the right, so the histograms of the two blurred images will be different.
(b) To handle the border effects, we surround the image with a border of 0s. We assume that image is of size N $\times$N (the fact that the image is square is evident from the right image in the problem statement). Blurring is implemented by a 3$\times$3 mask whose coefficients are 1/9. You can write a code to calculate and plot the histograms for both figures. Please note that different choices of N would lead to different answers for the right figure.
(c) A larger N would result in a larger number of 0/255 in the output histogram; while a smaller N would result in a smaller number of extreme values. But you should make sure that the summation of the histogram bins equals to N $\times$N.

2. Consider spatial filtering.
   (a) Suppose that you filter an image $f(x, y)$, with a spatial filter mask $w(x, y)$, using convolution, as defined in Eq. (3.4-2), where the mask is smaller than the image in both spatial directions. Show the important property that, if the coefficients of the mask sum to zero, then the sum of all the elements in the resulting convolution array (filtered image) will be zero also (you may ignore computational inaccuracies). Also, you may assume that the border of the image has been padded with the appropriate number of zeros. (10 pts)

$$w(x, y) \otimes f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x - s, y - t) \qquad (3.4 - 2)$$

   (b) Would the result to (a) be the same if the filtering is implemented using correlation, as defined in Eq. (3.4-1)? (10 pts)

$$w(x, y) \odot f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x + s, y + t) \qquad (3.4 - 1)$$

Solution:

(a) The key to solving this problem is to recognize (1) that the convolution result at any location (x,y) consists of centering the mask at that point and then forming the sum of the products of the mask coefficients with the corresponding pixels in the image; and (2) that convolution of the mask with the entire image results in every pixel in the image being visited only once by every element of the mask (i.e., every pixel is multiplied once by every coefficient of the mask). Because the coefficients of the mask sum to zero, this means that the sum of the products of the coefficients with the same pixel also sums to zero. Carrying out this argument for every pixel in the image leads to the conclusion that the sum of the elements of the convolution array also sum to zero.

(b) The only difference between convolution and correlation is that the mask is rotated by 180◦. This does not affect the conclusions reached in (a), so correlating an image with a mask whose coefficients sum to zero will produce a correlation image whose elements also sum to zero.

3. The three images shown were blurred using square averaging masks of sizes n=23, 25, and 45, respectively. The vertical bars on the left lower part of Figure 2(a) and (c) are blurred, but a clear separation exists between them. However, the bars have merged in image Figure 2 (b) in, in spite of the fact that the mask that produced this image is significantly smaller than the mask that produced image Figure 2 (c). Explain the reason for this. The vertical bars are 5 pixels wide, 100 pixels high, and their separation is 20 pixels. (Hints: you can simplify the problem by considering a single scan line through the bars in the image.) (20 pts)

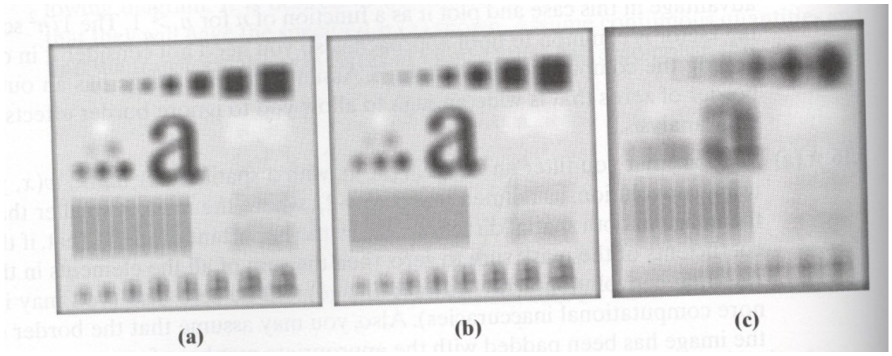
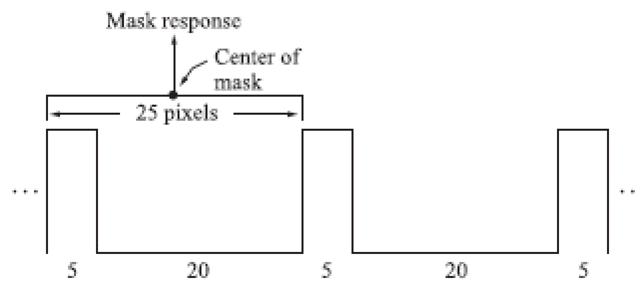


Figure 2. Three images smoothed by averaging using different filters

Solution:

We know that the vertical bars are 5 pixels wide, 100 pixels high, and their separation is 20 pixels. The phenomenon in question is related to the horizontal separation between bars, so we can simplify the problem by considering a single scan line through the bars in the image. The key to answering this question lies in the fact that the distance (in pixels) between the onset of one bar and the onset of the next one (say, to its right) is 25 pixels.

Consider the scan line shown in the figure shown below. Also shown is a cross section of a 25 $\times$25 mask. The response of the mask is the average of the pixels that it encompasses. We note that when the mask moves one pixel to the right, it loses one value of the vertical bar on the left, but it picks up an identical one on the right, so the response doesn't change. In fact, the number of pixels belonging to the vertical bars and contained within the mask does not change, regardless of where the mask is located (as long as it is contained within the bars, and not near the edges of the set of bars).



The fact that the number of bar pixels under the mask does not change is due to the particular separation between bars and the width of the lines in relation to the 25-pixel width of the mask. This constant response is the reason why no white gaps are seen in the image shown in the Fig. 1(b). Note that this constant response does not happen with the 23×23 or the 45×45 masks because they are not "synchronized" with the width of the bars and their separation.

4. In a given application an averaging mask is applied to input images to reduce noise, and then a Laplacian mask is applied to enhance small details. Would the result be the same if the order of these operations were reversed? (20 pts)

Solution:

Since both the Laplacian and the averaging process are linear operations, so it makes no difference which one is applied first.

5. Given a 3x3 filter shown in Figure 3 (a), perform the image filtering using **convolution** on
   (a) a 2x2 image as shown in Figure 3 (b) (10 pts)
   (b) a 3x3 image as shown in Figure 3 (c) (10 pts)
   Give the **FULL** filtering result for each filtering process. (Hint: you need to assume an appropriate zero mapping.)

-1	0	-1
0	2	0
1	0	1

(a)

1	2
3	2

(b)

0	0	0
0	1	0
0	0	0

(c)

Figure 3.

Solution:

(a)

-1	-2	-1	-2
-3	0	1	-2
1	8	5	2
3	2	3	2

(b) Using the property that convolution of a filter/mask with a discrete unit impulse copies the filter/mask at the location of the impulse.

0	0	0	0	0
0	-1	0	-1	0
0	0	2	0	0
0	1	0	1	0
0	0	0	0	0